

CAPSCAN



CSE 208: Database System Lab Project

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CapScan

CapScan is a smart tool that verifies the authenticity and actual capacity of memory cards and USB drives. It checks whether devices are genuine or not, measures read/write speeds, and uses a database for authenticity. CapScan helps prevent data loss and ensures secure, dependable storage, Ideal for students, photographers, and professionals.

Background:

CapScan is a simple and smart desktop application designed to check whether a memory card or USB drive is genuine and whether its actual storage capacity matches what is advertised.

Many storage devices, especially low-cost or counterfeit ones, often show a higher capacity than they truly have. This can lead to data loss and frustration when files suddenly disappear or become corrupted.

The tool provides clear, easy-to-understand results so users can quickly identify if their device is genuine, slow, or corrupted. It's especially useful for students, photographers, technicians, and anyone who regularly uses memory cards or flash drives.

Motivation:

- Many memory cards and USB drives display fake capacities or slow speeds, leading to data loss.
- Users need a simple and accurate tool to verify their storage devices.
- A tool like this isn't easily found or doesn't always work properly, leaving users without a dependable solution.
- CapScan provides a lightweight, cross-platform, and user-friendly solution for verifying drive authenticity, capacity, and performance.

Key Features

For Users:

- **Capacity Test:** Verifies the real usable capacity versus the advertised one.

- **Data Integrity Check:** Writes and reads test data to detect corruption or fake sectors.
- **Speed Test:** Measures read and write speeds.
- **Result Summary:** Displays tested capacity, errors, and actual usable size.
- **Export Results:** Save detailed test reports in TXT or CSV format.
- **Language Toggle:** Switch between English, Japanese or other languages.

For Device Inspectors:

- **Custom Test Range:** Option to test full or partial storage space.
- **Quick Test Mode:** Fast testing option for initial verification.
- **Detailed Logs:** Generates block-by-block data for advanced diagnostics.
- **Error Highlighting:** Marks corrupted or failed data.
- **Report Generation:** Export test summaries for record or customer proof.

Tools Used:

- **Programming Language:** Python
- **GUI Framework:** Tkinter
- **File I/O Operations:** Python's built-in I/O libraries
- **Performance Measurement:** OS-level read/write benchmarking
- **Database:** MySQL / SQLite for test history
- **IDE / Development Tool:** Visual Studio Code (VS Code)